

Mitigating Knowledge Conflicts of Retrieval-Augmented Generation through Dual-Stage Confidence Measurement in Semantic Space

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Abstract

While Retrieval-Augmented Generation (RAG) extends the knowledge boundaries of large language models (LLMs) by incorporating external evidence, it also introduces a critical challenge: the retrieved information may be irrelevant or misleading, and the model’s own parametric knowledge may be biased or outdated. As a result, RAG systems face two types of knowledge conflict: 1) Inter-context conflict, which arises when multiple retrieved documents contain mutually inconsistent information; and 2) Context-memory conflict, which occurs when the retrieved context contradicts the model’s internal parametric knowledge. Existing methods to solve knowledge conflicts often require additional training, exhibit limited generalization, and typically address these conflict types separately rather than within a unified framework. We propose Dual-Stage Confidence Measurement in Semantic Space (DS-CMS²), a unified training-free framework for mitigating both types of knowledge conflict simultaneously. DS-CMS² measures semantic confidence before and after inference, and uses it to adaptively filter input knowledge sources and adjust the model’s output semantic distribution. This dual-stage design enables flexible and principled integration of both internal parametric knowledge and external retrieved knowledge without requiring any additional training. Experiments on multiple benchmarks and LLMs demonstrate that DS-CMS² achieves strong effectiveness and robustness under diverse knowledge conflict settings. Code is provided at <https://github.com/hnnuzhangjia/CMS-2>.

1 Introduction

Large Language Models (LLMs) acquire substantial knowledge during pre-training, which is embedded within their parameters (Geva et al., 2021). However, constrained by the static nature of parametric knowledge, LLMs struggle to adapt over time and remain susceptible to hallucinations. Consequently, Retrieval-Augmented Generation (RAG), which involves gathering external and dynamic knowledge, has emerged as a standard paradigm for LLMs to bridge the temporal gap between training and inference (Guu et al., 2020; Lewis et al., 2020).

Despite these advances, recent studies reveal that retrieval augmentation carries the risk of introducing irrelevant or even contradictory knowledge (Xiang et al., 2024; Zou et al., 2024). Moreover, the knowledge embedded within LLMs may be biased or outdated due to inherent constraints in the training corpora (Bender et al., 2021; Navigli et al., 2023). Thus, RAG inevitably introduces two types of knowledge conflict: **1) Inter-context conflict:** Conflicts among different external contextual sources. **2) Context-memory conflict:** Conflicts between context and internal memory. As illustrated in Figure 1a, inter-context conflict arises

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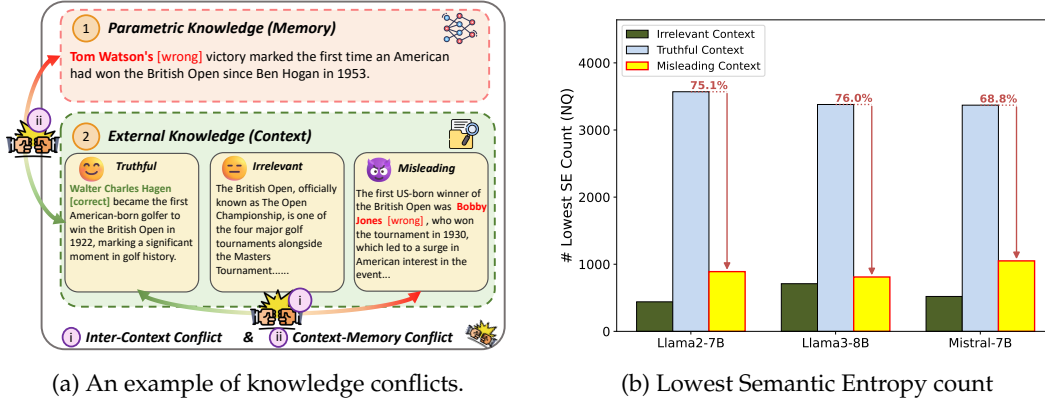


Figure 1: Illustration of knowledge conflicts and semantic entropy analysis.

when external knowledge contains a mixture of truthful (e.g., “Walter Charles Hagen”), irrelevant and misleading information (e.g., “Bobby Jones”). Moreover, context-memory conflict occurs when the model’s internal parametric memory (e.g., “Tom Watson”) diverges from the retrieved external knowledge (e.g., “Walter Charles Hagen”). These dual conflicts inject epistemic uncertainty into the RAG process, thereby undermining the reliability of the generated outputs. Therefore, formulating robust mechanisms to mitigate these knowledge conflicts is essential for the deployment of effective and trustworthy RAG systems.

Existing approaches to knowledge conflicts in RAG typically address inter-context conflict and context-memory conflict separately. For inter-context conflict, prior methods rely on either supervised training (Hsu et al., 2021; Hong et al., 2024) or fixed-strength contrastive decoding (Jin et al., 2024), which may generalize poorly or lead to overcorrection (Wang et al., 2025a). For context-memory conflict, fine-tuning methods such as KAFT (Li et al., 2023a) and DisentQA (Neeman et al., 2023) require additional training, while training-free decoding strategies such as CAD (Shi et al., 2024) and COIECD (Yuan et al., 2024) prioritize contextual information without assessing its reliability. As a result, existing methods still lack a unified framework for handling both conflict types simultaneously.

Recently, researchers found LLMs can produce diverse textual outputs that are often semantically equivalent despite surface-level differences. To quantify the uncertainty underlying such semantic variability, Kuhn et al. (2023) introduced semantic entropy, which measures uncertainty by clustering semantically equivalent generations sampled from a model and estimating entropy over the resulting meaning clusters. Unlike token-level probability metrics, semantic entropy captures uncertainty at the semantic level. Recent work has demonstrated its effectiveness in detecting hallucinations and assessing the reliability of LLM-generated responses (Farquhar et al., 2024; Kossen et al., 2024). Building on this line of research, we further analyze semantic entropy under different knowledge conditions and highlights our experiments (cf. Figure 1b, Figure 5) that it can serve as a useful signal for measuring semantic confidence and understanding knowledge conflicts in LLMs.

In this paper, we propose a novel training-free framework, **Dual-Stage Confidence Measurement in Semantic Space (DS-CMS²)**, which mitigates knowledge conflicts in RAG by measuring and adjusting the semantic confidence of the model. Specifically, our approach involves two key stages: **1) Pre-inference Knowledge Filtering via Semantic Confidence**. In this stage, we mitigate inter-context conflicts by filtering out irrelevant and misleading knowledge based on semantic confidence. Our experimental results (cf. Figure 1b, Figure 5) indicate that LLMs tend to exhibit lower semantic confidence when exposed to such unreliable information. Leveraging this observation, we compute semantic confidence to identify and eliminate untrustworthy knowledge sources and obtain reliable knowledge sources. **2) Post-inference Semantic Contrastive Decoding**. After obtaining reliable knowledge sources, we adjust the model’s semantic output distribution to mitigate context-memory conflict using semantic confidence. To achieve this, we introduce Semantic

Contrastive Decoding (SCD), a decoding strategy that samples from a contrastive semantic distribution, thereby amplifying differences in semantic probabilities with and without context. In addition, we design a dynamic comparison mechanism that adaptively models the degree of adjustment based on the model’s semantic confidence. When the model exhibits high confidence, which indicates strong internal certainty, we apply only a slight contrastive adjustment. Conversely, for low-confidence responses, the mechanism enforces a significant contrastive adjustment to align the output with reliable contextual information. This approach enables flexible integration of internal and external knowledge, resulting in outputs that are both contextually grounded and semantically accurate.

In summary, our contributions are as follows:

- We propose **DS-CMS²**, a novel framework to tackle inter-context conflict and context-memory conflict simultaneously through dual-stage confidence measurement in semantic space.
- We introduce a novel decoding method, named **SCD**, enabling the model to dynamically adjust the semantic distribution according to semantic confidence, making full use of the internal and external knowledge of the model.
- Extensive experiments demonstrate the effectiveness and robustness of our method in identifying the truthful knowledge sources and handling two types of knowledge conflicts. Notably, our method achieves an average improvement of 5.17% over the strongest baseline.

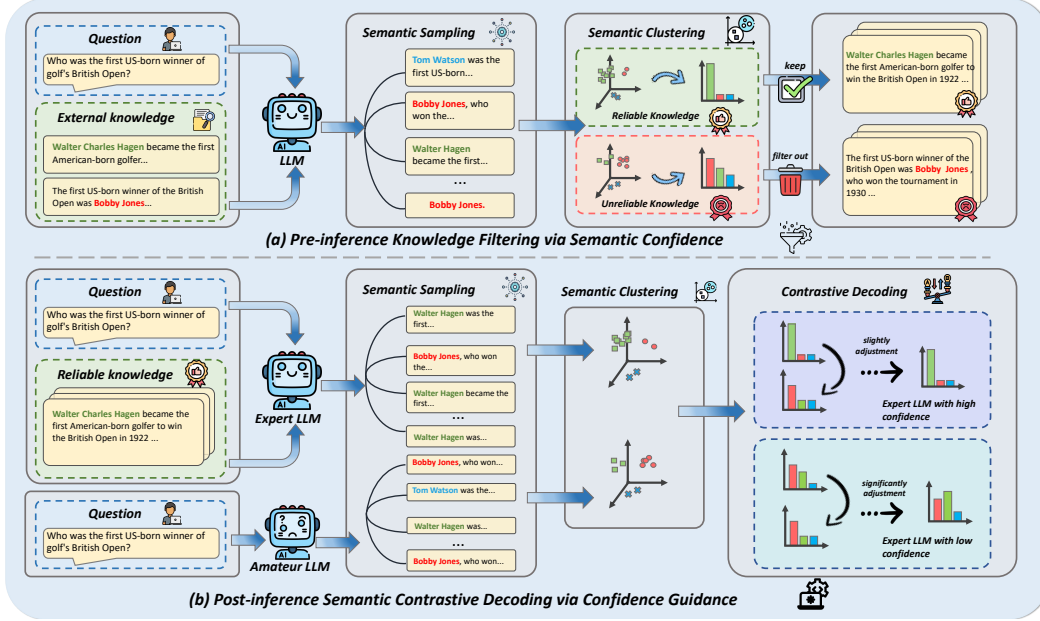
2 Related Work

2.1 Knowledge Conflicts

RAG has become a standard paradigm for knowledge-intensive tasks (Guu et al., 2020; Lewis et al., 2020), yet inevitably introduces inter-context conflict (Xiang et al., 2024; Zou et al., 2024) and context-memory conflict (Xu et al., 2024). Existing approaches address these conflicts in isolation. For inter-context conflict, methods rely on supervised training (Hsu et al., 2021; Hong et al., 2024) or knowledge selection (Wang et al., 2024; Kim et al., 2025; Wang et al., 2025b). For context-memory conflict, contrastive decoding methods such as CAD (Shi et al., 2024) and COIECD (Yuan et al., 2024) amplify differences between context-augmented and context-free outputs (Li et al., 2023b), but struggle when the compared distributions are highly similar, risking suppression of correct tokens (Wang et al., 2025a). Adaptive variants like ConfCD (Zhao et al., 2024) and AdaCAD (Wang et al., 2025a) modulate contrast strength based on confidence or divergence metrics, yet remain confined to token-level distributions and address only a single conflict type. In contrast, we propose a unified, training-free framework that simultaneously resolves both conflicts by operating in semantic space.

2.2 Semantic Entropy

LLMs exhibit remarkable diversity in their generative capabilities, which can lead to semantically identical outputs emerging from different generations. To address the challenge of quantifying semantic uncertainty in LLM-generated content, the concept of semantic entropy has been introduced (Kuhn et al., 2023). Semantic entropy measures uncertainty by clustering semantically equivalent sentences sampled from the model’s output. Recent studies have highlighted the effectiveness of semantic entropy in identifying and mitigating hallucinations in texts generated by LLMs (Farquhar et al., 2024; Kossen et al., 2024). Our experiments (cf. Figure 1b, Figure 5) reveal the differences in semantic entropy of large models when facing different types of knowledge sources, and we use it to measure semantic confidence and resolve knowledge conflicts at the semantic level.

Figure 2: The overall framework of our proposed DS-CMS².

3 Methodology

The framework of DS-CMS² is shown in Figure 2. Our framework has two components and is designed to flexibly integrate parametric and retrieved knowledge without additional training. In order to solve **inter-context conflict**, we first calculate the semantic confidence of the model under different knowledge sources and automatically keep reliable knowledge sources by setting an appropriate threshold. Furthermore, after obtaining the semantic distribution of the model inference, we dynamically adjust the semantic distribution of the model through the semantic confidence of the model output to achieve a comprehensive application of the internal and external knowledge and solve **context-memory conflict**.

3.1 Pre-inference Knowledge Filtering via Semantic Confidence

Semantic Confidence Measurement In order to overcome the problem of relying on internal representation (Azaria & Mitchell, 2023) in traditional confidence measurement, we use semantic entropy (Farquhar et al., 2024) to measure model confidence by calculating the entropy of the model’s semantic distribution. Specifically, low semantic entropy indicates that the model makes predictions with high confidence. In contrast, a high entropy value suggests that the model prediction is unstable and less reliable. We use semantic entropy to distinguish whether a knowledge source is reliable. The following sections describe the specific operation of using semantic entropy to perform knowledge filtering.

Semantic Sampling Given a knowledge source c_i , we feed c_i and x into the model and sample M sequences. The sampling temperature should be set high enough to sample a sufficient amount of semantics. The probability distribution of the sequence is as follows:

$$p_\theta(s | x, c_i) = \prod_{t=1}^n p_\theta(y_t | y_{<t}, x, c_i) \quad (1)$$

We sample sequences:

$$s_m \sim p_\theta(s | x, c_i), \quad m = 1, 2, \dots, M \quad (2)$$

Semantic Clustering and Semantic Entropy Computation. To compute semantic entropy, we first cluster model outputs that convey the same meaning. Following Farquhar et al. (2024), we leverage GPT-4o-mini (OpenAI, 2024) to determine whether two outputs are

semantically equivalent via bidirectional entailment. Semantically equivalent sequences are grouped into clusters $G = \{g_1, g_2, \dots, g_n\}$. We then estimate the semantic entropy over these clusters using the empirical distribution of sampled outputs:

$$H[G \mid x, c_i] \approx - \sum_{j=1}^n \frac{|g_j|}{M} \log \frac{|g_j|}{M} \quad (3)$$

Details of the clustering procedure and entropy derivation are provided in Appendix B.

Knowledge Filtering For each knowledge source c_i , we compute its semantic entropy $H[G \mid x, c_i]$ and select knowledge sources whose entropy does not exceed a threshold v as reliable. Formally, the set of reliable knowledge sources is

$$C_F = \{c_i \mid H[G \mid x, c_i] \leq v\} \quad (4)$$

If $C_F = \emptyset$, i.e., all retrieved knowledge sources exhibit semantic entropy above the threshold, we discard all external evidence and fall back to the closed-book setting, allowing the model to rely solely on its parametric memory.

3.2 Post-inference Semantic Contrastive Decoding

Adjustment in Semantic Space Given the computed C_F , we condition inference on (C_F, x) . However, due to the model’s reliance on parametric knowledge (Wan et al., 2023), its predictions may still exhibit incorrect semantics. This motivates the need to adjust the semantic distribution of the model’s inference outputs. SCD samples complete and coherent sequences from the model and selects as the final output the semantic class that exhibits the largest discrepancy between generations with and without context. Concretely, we instantiate two models via different prompts: an *expert* model (conditioned on context) and an *amateur* model (conditioned only on the question). From each model, we sample M sequences, denoted as S_{expert} and S_{amateur} . To ensure the semantic representations are strictly aligned for subtraction, we employ a **joint semantic clustering** strategy. We aggregate all sequences into a unified pool $S_{\text{total}} = S_{\text{expert}} \cup S_{\text{amateur}}$ and perform semantic clustering as described in Section 3.1. This yields a shared set of semantic classes $G = \{g_1, g_2, \dots, g_n\}$. For each semantic class $g_j \in G$, we calculate its frequency in the expert and amateur generations, denoted as $|e_j|$ and $|a_j|$, respectively. This unified space allows us to directly contrast the probability mass of identical meanings. Following Shi et al. (2024), we compute the final score for each semantic class g_j as:

$$\text{Score}(g_j) = \text{softmax}[(1 + \alpha)|e_j| - \alpha|a_j|] \quad (5)$$

After selecting the semantic class $g^* = \arg \max_j \text{Score}(g_j)$ with the highest contrastive score, we use the first sequence assigned to that class during clustering as the representative output, since it serves as the cluster’s anchor in our semantic equivalence procedure.

Dynamic α Previous contrastive methods (O’Neill et al., 2023; Pozzobon et al., 2023) use a fixed α for the entire dataset. However, every instance in the dataset may need a different weight for adjustment. In particular, models can sometimes generate relatively good answers without adjustment, and making a significant adjustment instead may harm the correct responses of the models (Wang et al., 2025a). To address the above problem, we dynamically set α for each instance based on the semantic confidence of the model. Intuitively, the optimal contrastive strength should be proportional to the model’s semantic uncertainty: when semantic entropy is low, the output already concentrates on a reliable semantic class, and aggressive adjustment risks overcorrection; when semantic entropy is high, the output is scattered across conflicting clusters, requiring stronger correction. Semantic entropy naturally satisfies this requirement—it is non-negative, self-bounding ($0 \leq H \leq \log n$), instance-adaptive, and operates in the same semantic space as our contrastive objective. We therefore directly adopt the semantic entropy of the expert model as α :

$$\alpha = - \sum_{j=1}^n \frac{|e_j|}{M} \log \frac{|e_j|}{M} \quad (6)$$

This ensures that the intervention strength is commensurate with conflict severity, avoiding both under-correction and over-correction.

4 Experiments

4.1 Experiment Setup

Datasets. Our evaluation framework spans two core tasks: Open-domain QA and Long-form QA. For Open-domain QA, we benchmark on NQ (Kwiatkowski et al., 2019), TriviaQA (Joshi et al., 2017) and WebQA (Berant et al., 2013). To probe long-form generation, we employ TruthfulQA (Lin et al., 2022), a dataset designed to challenge models with questions prone to eliciting false or misleading responses.

Metrics. We employ benchmark-specific metrics tailored to the nature of each task. For Open-domain QA, we report Span Exact Match (Span EM), which measures whether at least one ground-truth answer is contained within the model’s generated output. This differs from conventional exact match in that it assesses answer containment rather than strict string equivalence, making it more appropriate for evaluating the knowledge captured by LLMs. In the case of long-form QA, model performance is quantified using F1 and ROUGE-L scores.

Models. We test DS-CMS² on diverse pre-trained language models, including LLaMA2-7B (Touvron et al., 2023), LLaMA3-8B (AI@Meta, 2024) and Mistral-7B (Jiang et al., 2023).

Baselines. We adopt seven baselines organized into three categories. **Standard:** (1) Closed-book, where the model answers without external evidence; (2) Open-book, where all retrieved evidence is provided. **Decoding-based Strategies:** (3) CAD (Shi et al., 2024), which contrasts output distributions with and without context; (4) COIECD (Yuan et al., 2024), which uses information entropy to decide whether to apply contrastive decoding; (5) AdaCAD (Wang et al., 2025a), which adaptively calibrates contrast strength via Jensen-Shannon Divergence. **Adaptive Retrieval Strategies:** (6) CRAG (Yan et al., 2024), which evaluates and filters unreliable retrieved passages before generation; (7) Astute RAG (Wang et al., 2024), which iteratively consolidates internal and external knowledge to mitigate imperfect retrieval. Detailed configurations are provided in Appendix C.

4.2 Main Results

The main experimental results are presented in Table 1. (1) **Overall Analysis.** DS-CMS² outperforms existing strong methods across most datasets and LLMs, with particularly pronounced improvements on LLaMA-3-8B and Mistral-7B. For example, using LLaMA-3-8B, DS-CMS² achieves an average score of 0.531, surpassing AdaCAD by 6.0 percentage points and COIECD by 9.3 percentage points. Similarly, using Mistral-7B, DS-CMS² achieves an average of 0.486, surpassing the second-best method by 7.8 percentage points. These consistent improvements across models from diverse series and varying scales highlight the robustness and generalizability of our dual-stage semantic confidence framework. (2)

Impact of Model Ability. The advantage of DS-CMS² grows with model capability: on LLaMA-2-7B, it outperforms the second-best method by 1.7 percentage points on average, whereas the margins increase to 6.0 and 7.8 percentage points on LLaMA-3-8B and Mistral-7B, respectively. This suggests that stronger models possess more reliable parametric knowledge that DS-CMS² can better leverage through confidence-guided integration. (3) **Indiscriminate Context Use May Harm Performance.** Methods that unconditionally favor external knowledge, such as CAD, can degrade performance when retrieved documents are noisy. For instance, CAD scores 0.496 on NQ with Mistral-7B, 13.0 points below the closed-book baseline (0.626). More broadly, naively incorporating retrieval results can hurt capable models—using Mistral-7B, the open-book setting underperforms closed-book on NQ (0.572 vs. 0.626) and TriviaQA (0.425 vs. 0.587). DS-CMS² mitigates this issue through its confidence-aware mechanism, consistently outperforming both closed-book and open-book baselines across all settings.

LLM	Method	Open-Domain QA (Span EM)			Long-form QA (F1 / ROUGE-L)		Average
		NQ	TriviaQA	WebQA	TruthfulQA		
LLaMA-2-7B	Standard						
	Closed-book	0.407	0.243	0.371	0.170	0.135	0.265
	Open-book	0.440	0.444	0.431	0.188	0.163	0.333
	Decoding-based Strategies						
	CAD	<u>0.530</u>	<u>0.501</u>	0.556	0.315	0.252	<u>0.431</u>
	COIECD	0.401	0.300	0.417	<u>0.323</u>	<u>0.264</u>	0.346
	AdaCAD	0.501	0.478	0.545	0.181	0.143	0.370
	Adaptive Retrieval Strategies						
	CRAG	0.447	0.456	0.438	0.202	0.177	0.344
	Astute RAG	0.483	0.509	0.482	0.217	0.203	0.379
	Ours (DS-CMS²)	0.551	0.518	<u>0.548</u>	0.340	0.281	0.448
Mistral-7B	Standard						
	Closed-book	0.626	0.587	0.305	0.083	0.078	0.336
	Open-book	<u>0.572</u>	0.425	0.545	0.075	0.071	0.338
	Decoding-based Strategies						
	CAD	0.496	0.431	0.530	0.216	0.204	0.375
	COIECD	0.494	0.427	<u>0.575</u>	<u>0.276</u>	<u>0.266</u>	<u>0.408</u>
	AdaCAD	0.517	0.485	<u>0.564</u>	0.068	0.066	0.340
	Adaptive Retrieval Strategies						
	CRAG	0.429	0.473	0.472	0.145	0.127	0.329
	Astute RAG	0.557	0.643	0.500	0.102	0.093	0.379
	Ours (DS-CMS²)	0.665	<u>0.625</u>	0.587	0.283	0.271	0.486
LLaMA-3-8B	Standard						
	Closed-book	0.590	0.494	0.315	0.112	0.086	0.319
	Open-book	0.604	0.548	0.527	0.264	0.229	0.434
	Decoding-based Strategies						
	CAD	0.544	0.491	<u>0.621</u>	0.302	0.178	0.427
	COIECD	0.494	0.581	0.567	0.329	0.221	0.438
	AdaCAD	<u>0.665</u>	0.552	0.554	<u>0.342</u>	<u>0.244</u>	<u>0.471</u>
	Adaptive Retrieval Strategies						
	CRAG	0.621	0.572	0.598	0.174	0.168	0.427
	Astute RAG	0.572	0.541	0.484	0.168	0.159	0.385
	Ours (DS-CMS²)	0.689	0.601	0.637	0.375	0.355	0.531

Table 1: Evaluation results on Open-Domain QA (NQ, TriviaQA, WebQA) and Long-form QA (TruthfulQA), using LLaMA-2-7B-chat, Mistral-7B-Instruct and LLaMA-3-8B-Instruct as base LLMs. The best performance is marked in **bold**, and the second-best is underlined.

4.3 Fine-grained Evaluation

In this section, we design four evaluation scenarios to disentangle the effects of different conflict types: **No Conflict (NC)**: the model’s parametric memory encodes the correct answer and is consistent with the retrieved sources. **Inter-context Conflict (ICC)**: the model’s parametric memory encodes the correct answer, but retrieved sources contradict one another; **Context–Memory Conflict (CMC)**: parametric memory fails to produce the correct answer, while a reliable external source is available; **Both Conflict (Both)**: both ICC and CMC co-occur. Table 2 and Figure 3 report accuracy across all scenarios on NQ and WebQA. Several key findings emerge. (i) **Dynamic decoding strategies** (CAD, AdaCAD) exhibit limited robustness under conflict. On WebQA, CAD drops from 73.20% (NC) to 50.20% (ICC); AdaCAD degrades to 61.00% even without conflict (vs. 89.76% for ours), indicating that indiscriminate distribution adjustment introduces noise. On NQ, AdaCAD similarly falls from 75.80% to 40.80% when both conflicts co-occur. (ii) **CRAG** attains high ICC accuracy on both datasets (78.00% on NQ, 81.00% on WebQA), yet collapses under CMC (18.80% / 14.00%) and Both (18.20% / 16.40%). This reveals that its retrieval

Method	NQ					WebQA				
	NC	ICC	CMC	Both	Overall	NC	ICC	CMC	Both	Overall
CAD	72.80	49.60	<u>56.60</u>	<u>44.00</u>	55.75	73.20	50.20	57.40	39.00	54.95
COIECD	73.40	64.20	51.40	43.40	58.10	68.60	65.20	37.20	36.60	51.90
AdaCAD	<u>75.80</u>	<u>70.80</u>	48.40	40.80	<u>58.95</u>	61.00	65.20	28.00	28.60	45.70
CRAG	79.60	78.00	18.80	18.20	48.65	79.80	<u>81.00</u>	14.00	16.40	47.80
Astute RAG	61.70	69.20	55.00	38.00	55.98	<u>85.00</u>	81.40	<u>48.40</u>	29.00	<u>60.95</u>
Ours (DS-CMS²)	74.41	66.54	57.00	48.55	61.63	89.76	71.52	<u>55.60</u>	<u>36.92</u>	63.45

Table 2: Fine-grained evaluation (%) across conflict scenarios on NQ and WebQA using Mistral-7B. Overall is the average of the four scenario columns. The best performance is marked in **bold**, and the second-best is underlined.

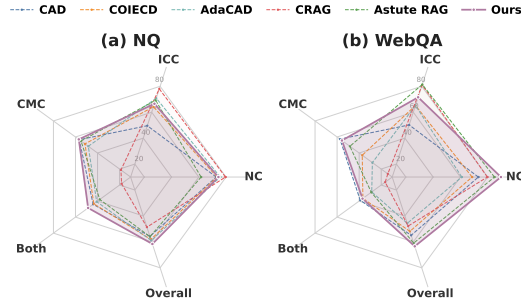


Figure 3: Radar-chart visualization of the fine-grained evaluation results in Table 2 across conflict scenarios on NQ and WebQA using Mistral-7B.

evaluator can detect inter-context inconsistency but fails when the model must selectively *absorb* external knowledge that contradicts parametric memory. (iii) **Our method** achieves the best overall accuracy on both NQ (61.63%) and WebQA (63.45%), with consistently competitive performance across all scenarios. In particular, it ranks first in CMC and Both on NQ (57.00%, 48.55%) and first in NC on WebQA (89.76%), demonstrating its ability to dynamically balance parametric and contextual knowledge without bias toward either source.

4.4 Ablation Study

To assess the core components of DS-CMS², we conduct a comprehensive ablation study on TriviaQA. As illustrated in Table 3, we evaluate the impact of the Semantic Contrastive Decoding (SCD) module by replacing it with alternative strategies: Open-book, CAD, COIECD, and AdaCAD. Additionally, we evaluate the necessity of the Knowledge Filtering (KF) module by removing it (w/o KF). A significant performance degradation is observed when SCD is replaced by other decoding methods, underscoring its critical role in our framework. By leveraging semantic contrastive decoding, the SCD module effectively integrates internal and external knowledge while dynamically calibrating the output distribution in the semantic space, thereby yielding more factually consistent results. The removal of KF also triggers a pronounced performance drop, particularly in larger-scale models. This phenomenon highlights the

Method	LLaMA2-7B	LLaMA3-8B	Mistral-7B
Open-book	0.464	0.613	0.589
CAD	0.422	0.543	0.494
COIECD	0.231	0.519	0.508
AdaCAD	0.482	0.619	0.563
w/o KF	0.430	0.485	0.461
DS-CMS²	0.518	0.625	0.601

Table 3: Ablation study on TriviaQA. The best performance is highlighted in **bold**.

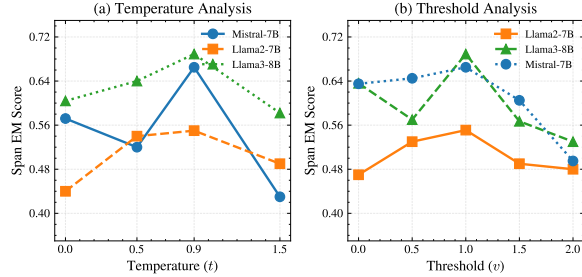
Method	Granularity	Framework Compat.	API Compat.	Avg. Latency (s/query)	Avg. Accuracy
Standard RAG	Sequence	✓	✓	0.14	0.434
CAD	Token	✗	✗	1.27	0.427
COIECD	Token	✗	✗	6.81	0.438
AdaCAD	Token	✗	✗	6.80	0.471
CRAG	Sequence	✓	✓	0.23	0.427
Astute RAG	Sequence	✓	✓	0.21	0.379
DS-CMS² (Ours)	Semantic	✓	✓	3.34 (two stages)	0.531

Table 4: Comparison of computational efficiency across different methods.

importance of KF in filtering out misleading and irrelevant information, thereby validating the necessity of reliable external knowledge for enhancing model accuracy.

4.5 Hyperparameter Analysis

Figure 4 illustrates the sensitivity of our framework to the sampling temperature t and knowledge filtering threshold v on NQ. For temperature t (Figure 4a), a moderate value is crucial for effective semantic sampling: an excessively low t yields nearly identical and deterministic outputs that hinder the semantic contrastive mechanism, whereas a high t produces highly unstable and noisy generations. Across all models, $t = 0.9$ achieves the optimal balance for semantic diversity. Similarly, the filtering threshold v (Figure 4b) exhibits an inverted U-shaped trend. An overly restrictive threshold (low v) erroneously discards essential truthful evidence, while excessively large values fail to filter out misleading noise from the context. With optimal performance consistently observed at $v = 1.0$, we adopt $t = 0.9$ and $v = 1.0$ as our default configurations to ensure robust semantic diversity and precise knowledge filtering.

Figure 4: The impact of t and v on our method across different models.

4.6 Computational Efficiency

DS-CMS² requires sampling M sequences per knowledge source, which may raise computational concerns. We measure wall-clock latency under identical conditions (vLLM serving Mistral-7B-Instruct). As shown in Table 4, DS-CMS² achieves the highest accuracy (0.531) while being **2.0 $\times$ faster than token-level methods** (AdaCAD/COIECD, 6.80–6.81 s), which require per-step logit intervention that disables KV-cache optimization and precludes framework/API compatibility. Our semantic-level design leverages vLLM’s KV-cache reuse to generate $M=10$ samples in parallel, with self-verification adding only $O(M)$ judgments ($<15\%$ overhead). Although lightweight baselines (CRAG, Astute RAG, CAD) are faster, they suffer substantial accuracy degradation under conflicts (0.379–0.427; cf. Table 4). Overall, DS-CMS² attains the best accuracy–latency trade-off.

5 Conclusions and Limitations

We present **DS-CMS²**, a training-free framework that mitigates knowledge conflicts in RAG by operating directly in semantic space. The framework addresses two coexisting conflict types through dual-stage confidence measurement: a pre-inference knowledge filtering stage that identifies and discards unreliable evidence via semantic entropy, and a post-inference semantic contrastive decoding stage that dynamically balances internal and external knowledge through a confidence-adaptive adjustment coefficient α . Experiments

across four benchmarks and three LLMs demonstrate consistent improvements over strong baselines. The semantic-level design further confers black-box API compatibility and favorable inference latency compared to token-level alternatives. A limitation of DS-CMS² is that the dynamic α may be small when the context-conditioned model is highly confident. Although such confidently wrong cases are relatively rare in practice (Appendix E), they may still lead to insufficient correction toward retrieved evidence when the context-conditioned model remains highly confident.

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A Prompt Examples

A.1 Knowledge Source Construction

To ensure ecological validity while guaranteeing that the test data contains diverse types of knowledge conflicts, we adopt a **hybrid strategy** that combines real retrieval with controlled injection. This approach prioritizes naturally occurring conflicts from real retrieval results and supplements with synthetic generation only when natural conflicts are absent.

Step 1: Real Retrieval. For each question x in the dataset, we employ Contriever as the retrieval backbone to retrieve the top- K ($K=3$) passages from the English Wikipedia dump. This yields a candidate passage set $\mathcal{P} = \{p_1, p_2, p_3\}$ that reflects realistic retrieval noise, including naturally occurring irrelevant or contradictory information.

Step 2: Conflict Detection. We then assess whether the retrieved passages exhibit sufficient knowledge conflicts. Specifically, we perform two checks:

- **Inter-context conflict check:** We use a DeBERTa-based NLI model (He et al., 2020) to perform pairwise entailment detection among the retrieved passages with respect to the question x . If any pair (p_i, p_j) yields a *contradiction* judgment, we consider inter-context conflict to be present.
- **Answer consistency check:** We verify whether the passages support divergent answers by prompting the base LLM to answer x conditioned on each passage independently. If the induced answers disagree, this further confirms the presence of conflict.

Step 3: Conditional Augmentation. Based on the conflict detection results, we apply one of two strategies:

- **If conflicts are detected** ($\exists (p_i, p_j)$ with contradiction or divergent answers): We directly adopt the retrieved passages $\mathcal{P}$ as the external knowledge sources. These naturally conflicting passages preserve realistic noise characteristics, including subtle misleading cues and genuine irrelevance, thereby ensuring high ecological validity.
- **If no conflict is detected** (all passages are consistent): We retain the most relevant retrieved passage as the truthful evidence and use GPT-4o-mini to generate supplementary misleading and irrelevant passages, ensuring that the evaluation covers conflict scenarios. The generation prompt is provided below.

Prompt for Supplementary Knowledge Generation

Given the following question, answer, and a truthful evidence passage retrieved from Wikipedia, generate two additional passages:

1. **wrong_evidence:** A passage that appears plausible and well-written but leads to an incorrect answer. It should contradict the truthful evidence while remaining internally coherent.
2. **irrelevant_evidence:** A passage that is topically related to the question’s domain but does not contain information useful for answering the question.

Truthful evidence: {retrieved_passage}

Question: {question}

Answer: {answer}

Following is an example of the expected output format:

{“wrong_evidence”: “Zodiac signs are determined by the position of the sun at the time of a person’s birth. Given Jamie Lee Curtis’s birth date, she is officially classified under the Sagittarius sign.”,

“irrelevant_evidence”: “Jamie Lee Curtis is a famous American actress, writer and producer, renowned for her outstanding performances in the fields of film and television.”}

A.2 Model Setup Prompt

In order to achieve semantic comparison, we input contextual knowledge into the expert model and set prompt words to enhance the model’s capabilities. For amateur models, we do not input any context into the model and set prompt words that weaken the model’s ability.

Prompt for expert model

```
<user>
You are a professional question-answering model that can determine whether the
provided evidences is correct and give the correct answer to the question.
This evidence includes the memory that is related to the answer, irrelevant or incorrect
evidences:context
you must Answer the following question in a single brief but complete sentence.
Question: question
Answer:
```

Prompt for amateur model

```
<user>
You are an amateur in question answering.
you must Answer the following question in a single brief but complete sentence.
Question: question
Answer:
```

A.3 Semantic Equivalence Judge and Prompt

To perform semantic clustering, we require a semantic equivalence function that determines whether two sampled outputs belong to the same meaning cluster. Following Farquhar et al. (2024), we employ GPT-4o-mini as the default semantic equivalence module due to its strong semantic matching capability. Importantly, DS-CMS² only assumes access to a semantic equivalence function, making the backend modular and replaceable.

Below we provide the prompt used for GPT-4o-mini:

Prompt for Semantic Equivalence Judge

```
<user>
We are evaluating answers to the question: {question}. Here are two possible
answers:
Possible Answer 1: {text1}
Possible Answer List 2: {text2}
Does Possible Answer 1 semantically entail any element in Answer List 2? Your
response must be entailment, contradiction, or neutral.
Answer:
```

A.4 Open-source Semantic Equivalence Backend

To evaluate whether GPT-4o-mini is essential to our framework, we replace it with a locally deployed **DeBERTa-v3-large** (cross-encoder/nli-deberta-v3-large) while keeping all other components unchanged.

Table 5 compares the two semantic equivalence backends. Replacing GPT-4o-mini with DeBERTa-v3-large results in only a modest performance drop (approximately 1.5% ~ 3.6% on average), while DS-CMS² consistently remains stronger than all baselines reported in the main paper. We further verify that both the backbone LLM and DeBERTa-v3-large can be deployed on a single NVIDIA A800 GPU. Under batched inference, the average end-to-end latency increases moderately from 3.34 s to 4.56 s per query, eliminating external API dependency while maintaining competitive efficiency.

These results indicate that DS-CMS² is not tied to any specific semantic equivalence model. Practitioners may choose GPT-4o-mini for maximum accuracy or DeBERTa-v3-large for fully local deployment and reproducibility.

Table 5: Performance comparison between GPT-4o-mini and DeBERTa-v3-large as the semantic equivalence backend.

Model	Backend	NQ	TriviaQA	WebQA	TruthfulQA (F1/R-L)	Avg
LLaMA2-7B	GPT-4o-mini	0.551	0.518	0.548	0.340 / 0.281	0.448
	DeBERTa-v3	0.550	0.517	0.553	0.274 / 0.257	0.430
LLaMA3-8B	GPT-4o-mini	0.689	0.601	0.637	0.375 / 0.355	0.531
	DeBERTa-v3	0.674	0.574	0.592	0.330 / 0.305	0.495
Mistral-7B	GPT-4o-mini	0.665	0.625	0.587	0.283 / 0.271	0.486
	DeBERTa-v3	0.557	0.603	0.599	0.320 / 0.295	0.475

A.5 Baseline Prompt

Table 6 lists the prompts used during evaluation. We utilized the same prompts for all the baselines. The other baselines use the prompts set in the original paper.

Task Type	Prompt
<i>Closed-Book</i>	
Open-Domain QA (Natural Questions, TriviaQA, WebQA) Long-form QA (TruthfulQA)	You are a helpful assistant. Answer the following question concisely. Question: $\{question\}$? The answer is:
<i>Open-Book</i>	
Open-Domain QA (Natural Questions, TriviaQA, WebQA) Long-form QA (TruthfulQA)	You are a helpful assistant. Answer the following question based on the provided Background. Background: $\{background\}$ Question: $\{question\}$? Please analyze the evidence and provide your answer to the question. The answer is:

Table 6: Prompts used for different task types under Closed-Book and Open-Book settings.

B Semantic Clustering and Entropy Computation Details

Semantic Clustering. To compute semantic entropy, we first cluster model outputs that convey the same meaning. Following (Farquhar et al., 2024), we leverage natural language inference (NLI) models to estimate entailment relationships between output pairs. Formally, given two sequences s_1, s_2 , we consider them semantically equivalent if s_1 entails s_2 and s_2 entails s_1 . In this work, we adopt GPT-4o-mini as an evaluator L to determine whether two outputs s_1 and s_2 are semantically equivalent with respect to a given input x :

$$s_1 \equiv s_2 \iff L(x, s_1, s_2) = \text{True}. \quad (7)$$

We then perform semantic clustering based on semantic equivalence. Let s be a sequence and $G = \{g_1, g_2, \dots, g_n\}$ denote the current set of semantic classes. We assign s to an existing class $g_j \in G$ if it is semantically equivalent to at least one sequence in g_j . For efficiency, we compare s only with a representative sequence $s_{g_j,1}$, chosen as the first sequence added to g_j . If s is not semantically equivalent to any representative in G , we create a new semantic class g_{n+1} and add it to G .

Semantic Entropy Computation. The probability of a semantic cluster g_j is defined as the sum of the probabilities of all sequences belonging to that cluster:

$$p_\theta(g_j | x, c_i) = \sum_{s \in g_j} p_\theta(s | x, c_i). \quad (8)$$

Semantic entropy measures the uncertainty over semantic clusters and is computed as:

$$H[G | x, c_i] = - \sum_{j=1}^n p_\theta(g_j | x, c_i) \log p_\theta(g_j | x, c_i). \quad (9)$$

In practice, the exact semantic cluster probabilities are intractable due to the exponentially large output space. Following (Farquhar et al., 2024), we adopt a discrete approximation of semantic entropy based on sampling. Specifically, we treat the semantic clusters induced by sampled outputs as the full set of semantic clusters and estimate the entropy using their empirical probabilities:

$$p_\theta(g_j | x, c_i) \approx \frac{|g_j|}{M}, \quad (10)$$

where $|g_j|$ denotes the number of sampled sequences in cluster g_j and M is the total number of samples. The semantic entropy can then be approximated as:

$$H[G | x, c_i] \approx - \sum_{j=1}^n \frac{|g_j|}{M} \log \frac{|g_j|}{M}. \quad (11)$$

C Baselines and Dataset Details

C.1 Baselines

We adopt seven methods as baselines. Our experiment follows commonly used configurations found in recent research.

- (1) **Close-book:** The model answers questions without access to any external evidence. We set the temperature t to 0.1 to obtain stable outputs.
- (2) **Open-book:** The model answers questions based on the provided external context. We set the temperature t to 0.1 to obtain stable outputs.
- (3) **CAD:** which contrasts the output distributions of the model with and without contextual information to guide decoding. We set the contrast coefficient α to 1 and set the temperature t to 0.1.
- (4) **COIECD:** which leverages information entropy to assess the reliability of context and determines whether contrastive decoding should be applied. The values of λ and α are set to 0.25 and 1 respectively, and the temperature t is set to 0.1.
- (5) **AdaCAD:** which adaptively calibrates the degree of adjustment based on the Jensen-Shannon Divergence (JSD) between output distributions. AdaCAD employs a dynamic α .
- (6) **CRAG:** which assesses the quality of each retrieved document using a lightweight retrieval evaluator and applies corrective actions—such as filtering unreliable passages or triggering web search for supplementary information—before feeding the refined context to the generator.
- (7) **Astute RAG:** which mitigates the effects of imperfect retrieval by iteratively consolidating the model’s parametric knowledge with external evidence, thereby reducing conflicts between internal memory and retrieved context before generation.

For our proposed method, we set the semantic entropy threshold v to 1, and follow prior work to set the number of samples M to 10. We set the temperature t to 0.9 to sample richer semantics.

C.2 Dataset Details

Table 7 summarizes the dataset statistics used in our experiments. For Open-domain QA benchmarks (NQ, TriviaQA, and WebQA), we randomly sample 817 questions from the original test set and construct knowledge conflict scenarios following the procedure described in Appendix A.1. For TruthfulQA, we use the full dataset (817 questions) given its moderate size. The sampling strategy is motivated by the computational overhead of our method, which requires $M = 10$ semantic samples per knowledge source per query during both the pre-inference filtering and post-inference decoding stages.

Table 7: Dataset statistics used in our experiments.

Task	Dataset	Original Test Size	Used Size	Sampling
Open-domain QA	NQ	3,610	817	Random
	TriviaQA	11,313	817	Random
	WebQA	2,032	817	Random
Long-form QA	TruthfulQA	817	817	Full

D Effectiveness of Semantic Entropy

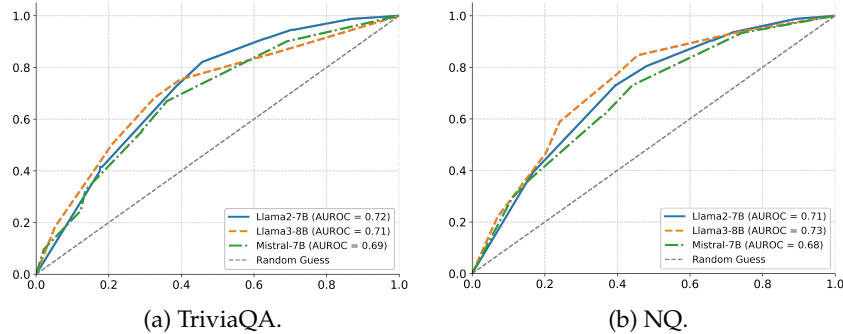


Figure 5: ROC curves for distinguishing truthful from misleading knowledge via semantic entropy.

As shown in Figure 5, we conducted experiments using three models in TriviaQA and NQ and plotted their Receiver Operating Characteristic (ROC) curves. Our analysis primarily focuses on the models’ ability to distinguish truthful knowledge from misleading knowledge. We calculate the AUROC and find that it is significantly higher than 0.5, further demonstrating the effectiveness of semantic entropy in identifying reliable knowledge sources.

E Frequency of “Confidently Wrong” Failure Mode

A potential concern is that semantic entropy may occasionally assign low uncertainty to an incorrect answer, resulting in a “confidently wrong” (CW) prediction. We define CW instances as cases where the semantic entropy falls below the filtering threshold used in DS-CMS² while the generated answer is incorrect. Table 8 reports the proportion of CW instances on three open-domain QA benchmarks. Across all backbone models, the CW ratio remains consistently below 6%, indicating that highly confident yet incorrect semantic distributions occur only infrequently. Moreover, even when such cases arise, the proposed Knowledge Filtering (KF) stage provides additional robustness by filtering unreliable external knowledge independently of the semantic entropy score. As a result, reliable evidence is often preserved for the subsequent Semantic Contrastive Decoding (SCD) stage, reducing the impact of occasional confidently wrong predictions.

Table 8: Proportion of “confidently wrong” (CW) instances. A CW instance is defined as an incorrect prediction whose semantic entropy is below the filtering threshold used in DS-CMS².

Model	TriviaQA	NQ	WebQA
LLaMA2-7B	4.55%	3.04%	5.11%
LLaMA3-8B	2.76%	1.02%	3.60%
Mistral-7B	3.21%	2.77%	4.34%

Setup	Both	CMC	ICC	Avg
$\alpha=0$	0.545	0.515	0.840	0.633
$\alpha=0.5$	0.598	0.616	0.880	0.698
$\alpha=1.0$	0.609	0.630	0.901	0.715
$\alpha=1.5$	0.615	0.618	0.900	0.712
dynamic α	0.651	0.639	0.900	0.730

Table 9: Accuracy of different α settings on context-memory conflict data (Cmc), inter-context conflict data (Icc) and data with both conflicts (Both). The term “Avg” refers to average. The best performance is highlighted in **bold**.

F Effectiveness of Dynamic α

Our method dynamically adjusts the value of α based on the semantic confidence of the model. To validate the effectiveness of this dynamic α configuration, we conducted comparative experiments using LLaMA3-8B against variants with fixed α on three datasets with different types of knowledge conflicts on TriviaQA. As illustrated in Table 9, while fixed α settings yield performance comparable to the dynamic approach on specific subsets, they consistently underperform the dynamic α across the majority of benchmarks and in terms of mean accuracy.

G Case Study

We use a concrete example to illustrate the advantages of our approach. It can be seen that the use of CAD and COIECD has damaged the decoding quality of the model, while AdaCAD has not declined in decoding quality due to the use of dynamic adjustment. Moreover, since these decoding methods do not filter external knowledge sources, the decoding methods are ultimately interfered with by erroneous information, resulting in the generation of incorrect answers.

Example

Question: What is the fastest animal on two legs?

External sources:

1.The ostrich is known to be the fastest bird on land, capable of running at speeds up to 45 miles per hour (72 km/h) over short distances and maintaining a speed of 30 to 40 miles per hour (48 to 64 km/h) over longer distances, making it the fastest two-legged animal.

2.The fastest animal on two legs is the emu, which can run at speeds of up to 30 miles per hour (48 km/h) but is significantly slower than the ostrich, typically maintaining a speed of just 15 to 20 miles per hour.

3.Ostriches are large flightless birds native to Africa, known for their long necks and legs, and they can weigh up to 290 pounds (130 kg).

Answer: ostrich

Close-book:Cheetah

Open-book:The cheetah is the fastest animal on two legs.

CAD:

COIECD: \"/nAnswer: The fastest animal on two legs is the cheetah

AdaCAD:The fastest animal on two legs is the cheetah, which can run up to 70 miles per hour (113 km/h).

CRAG:The cheetah is the fastest animal on two legs.

Astute RAG:based on the most reliable cluster, the answer to the query is ostrich.

DS-CMS²: The ostrich is the fastest bird on land and the fastest two-legged animal.